Spec Approvals

Multi-Level Approval Workflow for Vendor Bills & Purchase Orders

Kebony Inc. — Odoo 19 Custom Module

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Overview — What & Why

A unified approval layer for financial documents

Business Need

- All **Vendor Bills** and **Purchase Orders** must be approved before posting/confirmation
- Approval chain driven by the **Business Unit** assigned to each vendor
- Optional **amount-based thresholds** trigger additional approvers
- Full audit trail with email notifications at each step

Scope

- `account.move` — Vendor Bills
- `purchase.order` — Purchase Orders
- New `kebony.approval` configuration model
- Centralized approval list view & follow-up widget

Key Principles

Vendor-Driven Routing

Each vendor has a Business Unit. The BU determines which approval rule applies.

Sequential Approval

Approver 1 must sign off before Approver 2 is notified. No parallel approval.

Blocking Workflow

Documents cannot be posted/confirmed until all required approvers have approved.

Rejection Loop

Rejected documents return to the responsible user for corrections, then re-enter the approval chain.

Vendor Business Units

Every vendor must have a Business Unit assigned on their partner record

Business Unit Values

BUSINESS UNIT	DESCRIPTION	EXAMPLE VENDORS
Marketing	Agencies, events, promotional materials	Ad agencies, event suppliers
Production	Manufacturing and production suppliers	Raw material, chemicals
Logistics	Transport, freight, shipping	Carriers, freight forwarders
Warehousing	Storage and warehouse services	3PL providers, storage

Important: Business Units may differ depending on the supplier. This means we need **several Approval rules per model** — one per Business Unit configuration.

Vendor Form Configuration

New Fields on `res.partner`

- **Supplier Category:** Marketing, Production, Logistics, Warehousing
- **Supplier Dept:** Linked Business Unit selection
- **Business Unit:** Required field for approval routing

These fields appear on the *Sales & Purchases* tab of the vendor form.

Prerequisite: Vendors need to have a destination "Business Unit" before any approval workflow can route correctly.

New Approvals Model / App

`kebony.approval` — Central configuration for all approval rules

Core Configuration Fields

FIELD	DESCRIPTION
Approval name	Descriptive label for the rule
Model	`account.move` or `purchase.order`
Company	Scoped to a specific company
Approver 1	First (mandatory) approver
Approver 2	Second approver (conditional)
Business Unit	Links rule to vendor's BU
Business Unit Head	Auto-resolved from BU config

Amount-Based Extra Approver

`Approver extra`

- **Based on amount** — checkbox to enable threshold
- **Min** — minimum amount (currency) triggering extra approval
- **Amount approver** — user who must approve above threshold

Min & Amount approver fields only displayed if "Based on amount" is checked.

Model Selection Logic

```
Select between:
  ├─ account.move     (Vendor Bill approvals)
  │   └─ for specific vendor
  │       Company / BU combo
  │
  └─ purchase.order   (PO approvals)
      └─ for specific vendor
          Company / BU combo

Select any other model :
  └─ Custom fields
```

Validation Rules

Vendor Bills (`account.move`)

Always 2 validators: BU Head + another BU unit head (e.g., Finance)

Purchase Orders (`purchase.order`)

- **1 validator** if one → PO responsible only
- **2 validators** if no → BU Head + another BU head (e.g., Finance)

Workflow — Vendor Bills

`account.move` approval lifecycle

Step-by-Step Flow

Step 1 — User Confirms the Bill

- User opens Vendor Bill in `Draft` status
- Clicks **Confirm**
- System sends email notification to **Approver 1**
- Status → Waiting for Approval

↓

Step 2 — Approver Reviews

- Approver sees **Approve** and **Reject** buttons
- **Approve** → if Approver 2 exists, notification sent to next approver
- **Reject** → email sent to Bill responsible to make changes

↓

Step 3 — Final Approval & Posting

- Final approver clicks Approve → status changes to Approved
- Users from **"Accounting Department"** group can then **post** the invoice

Status Values

STATUS	MEANING
Draft	Initial state — editable
Waiting for Approval	Confirmed, pending approver action
Approved	All approvers signed off
Approval Rejected	Returned for changes
Posted	Standard Odoo posted state

Approval Flow Diagram

```
User clicks Confirm
  │
  ▼
Approver 1 receives notification
  ├── Approve ──▶ Approver 2 notified
  │                  ├── Approve ──▶ ✓ Approved
  │                  └── Reject  ──▶ Back to responsible
  └── Reject  ──▶ Back to responsible

After approval:
Accounting Dept ──▶ Post Invoice
```

Rule: All vendor bills must be approved by their respective BU head. Only **Accounting Department** users can post after approval.

Invoice Validation — PO Deviation Control

Strengthening 3-way matching with threshold-based deviation checks

Why Standard 3-Way Matching Falls Short

Odoo Native Behaviour

Standard 3-way matching (PO → Receipt → Invoice) only checks that a receipt exists. It does **not** flag price or quantity deviations between the PO and the vendor bill.

The Risk

- Vendor invoices higher unit prices than agreed on PO
- Invoiced quantities exceed received quantities
- Additional charges or fees not on the original PO
- All pass through without detection under standard matching

Proposed Enhancement

Deviation Check Before Approval

When a vendor bill is confirmed, the system compares each bill line against the linked PO line and flags deviations exceeding configurable thresholds.

Threshold Configuration

CHECK	COMPARISON	THRESHOLD
Unit Price	Bill line price vs PO line price	% tolerance (e.g., 5%)
Quantity	Bill qty vs received qty	% tolerance (e.g., 10%)
Line Total	Bill line amount vs PO line amount	Absolute & % tolerance
Unmatched Lines	Bill lines with no PO reference	Always flagged

Workflow Integration

```
Bill Confirmed
  │
  ▼
Deviation Check
  ├── Within tolerance ──▶ Normal approval flow
  │
  └── Exceeds tolerance ──▶ Flagged for review
        └── Deviation details visible
            in Approvals tab
            └── Approver must Accept
                or Reject deviation
```

Key rule: Bills with deviations above threshold **cannot** be approved without explicit acknowledgement of the deviation by the approver.

Workflow — Purchase Orders

`purchase.order` approval lifecycle

Step-by-Step Flow

Step 1 — User Confirms the PO

- User opens RFQ / Purchase Order
- Clicks **Confirm**
- System sends email notification to **Approver 1**
- Status → Waiting for Approval

Step 2 — Approver Reviews

- Approver sees **Approve** and **Reject** buttons
- **Approve** → if Approver 2 exists, notification sent to next approver
- **Reject** → email sent to PO responsible to make changes

Step 3 — Final Approval

- Final approver clicks Approve → status changes to Approved
- Users from **"PO Department"** group can then proceed with the Purchase Order

PO-Specific: If only 1 validator is configured, only the PO responsible approves. With 2 validators, the BU Head + another BU head (e.g., Finance) must both approve.

Status Values

STATUS	MEANING
Draft	Initial state (RFQ)
Waiting for Approval	Confirmed, pending approval
Approved	All approvers signed off
Approval Rejected	Returned for changes
Purchase Order	Standard Odoo confirmed PO

PO Approval Flow

```
User clicks Confirm
  │
  ▼
Approver 1 receives notification
  ├── Approve ──▶ Approver 2 (if exists)
  │                   ├── Approve ──▶ ✓ Approved
  │                   └── Reject  ──▶ Back to responsible
  └── Reject  ──▶ Back to responsible

After approval:
PO Department ──▶ Confirm PO
```

Conditional Validators

CONDITION	VALIDATORS
PO linked to a single responsible	**1** — PO responsible only
No single responsible / complex	**2** — BU Head + Finance (or other BU Head)

Approvals Follow-Up & UI

Embedded widgets, list views, and tracking

Approvals Tab (on Documents)

A new **"Approvals"** tab is added to both Vendor Bill and Purchase Order forms, alongside Invoice Lines, Journal Items, and Other Info.

This tab shows:

- Current approval status & history
- Which approver is next in line
- Timestamps of each approval/rejection action

Follow-Up Widget

Embedded in Document Form

- **Approver 1 / Approver 2** tabs showing individual status
- **Status** color-coded progress bar
- Created as a **sub-task** linked to the document
- **Statusbar widget** for visual progression

Tags for Tracking

TAG	COLOR	MEANING
Waiting for Approval	Orange	Pending approver action
Approved	Green	Approver signed off
Rejected	Red	Approver rejected

Centralized Approval List View

All requested approvals grouped in a single list, sortable and filterable:

COLUMN	DESCRIPTION
Date	Approval request date
Delay	Time elapsed since request
Approver	Assigned approver
Supplier	Vendor on the document
Model	Bill or PO
Business Unit	Vendor's BU
Amount	Document total amount
Status	Current approval status

> **Dynamic fields:** Additional columns adapt based on the Model — Vendor Bill-specific fields vs Purchase Order-specific fields.

Invoice Tracking Dashboard

SUIVI DES FACTURES

- **Time filters:** 1 Jour, 1 Mois, 2 Mois
- **Date range:** Custom period selector
- **Reminders:** Overdue approval tracking
- **Maturity:** Payment maturity overview

Notifications & Security

Email workflow, access rights, and audit trail

Notification System

On Confirm (User → Approver 1)

Email sent to Approver 1 with document link. Chatter message logged.

On Approve (Approver 1 → Approver 2)

If first approver clicks "Approve" and Approver 2 exists, an email/notification is sent to the next approver.

On Reject (Approver → Responsible)

An email/notification is sent back to the Bill/PO responsible so they can make changes and re-submit.

On Final Approval

Status changes to "Approved". Accounting / PO Department users can then post or confirm.

Security & Access Control

ROLE	CAN DO
Bill/PO Creator	Create, edit draft, click Confirm
Approver (current)	See Approve/Reject buttons (only when it's their turn)
Approver (not current)	View only — no action buttons
Accounting Department	Post approved Vendor Bills
PO Department	Confirm approved Purchase Orders

Audit Trail

- All approval actions logged with **timestamp + user**
- Chatter messages on each state transition
- Rejection reason tracked for accountability
- Full history visible in the Approvals tab

Key design: Approvers only see Approve/Reject buttons when it is specifically their turn in the chain. This prevents out-of-order approvals.

Budget Control Module

Linked workstream — developed by VVF in sync with the approval workflow

Why a Dedicated Module

Odoo Native Limitations

Odoo's built-in budget features are basic (analytic-account level only) and do not integrate with manufacturing cost allocation or inventory valuation standards.

Accounting Standards Requirement

Under **IAS 2 / ASC 330**, manufacturing indirect costs (overhead) must be capitalized into inventory. Budget control and cost allocation are two sides of the same coin — they must use the same cost pools and rates.

Scope — What the Module Covers

- **Budget definition** per cost centre / department / BU
- **Budget consumption tracking** against POs and vendor bills
- **Threshold alerts** when spend approaches budget limits
- **Integration with approval workflow** — over-budget POs/bills flagged during approval

Link to Manufacturing Indirect Costs

```
Budget Pools
  ├── Direct costs       (raw material, chemicals)
  │    └── tracked via PO/Bill
  │
  └── Indirect costs     (overhead, utilities, maint.)
       ├── Budget module tracks spend
       └── Same rates feed inventory
            valuation (IAS 2 / ASC 330)
```

One source of truth: Indirect cost rates defined in the budget module are the same rates used for manufacturing overhead absorption into inventory. No duplication, no drift.

Responsibility Split

COMPONENT	DEVELOPED BY
Approval workflow (this deck)	**Odoo** (custom dev)
PO deviation checks	**Odoo** (custom dev)
Budget control module	**VVF Consulting**
Manufacturing cost allocation	**VVF Consulting**

Sequencing: The budget module ships alongside or shortly after the approval workflow, so that budget checks can integrate into the approval chain from day one.

Implementation Summary

Technical components, data model, and responsibility split

NEW MODELS

`kebony.approval`

- Approval name
- Model (selection)
- Company (many2one)
- Approver 1 (many2one)
- Approver 2 (many2one)
- Business Unit (selection)
- Business Unit Head (related)
- Based on amount (boolean)
- Min amount (monetary)
- Amount approver (many2one)

`kebony.approval.deviation`

- Bill line (many2one)
- PO line (many2one)
- Deviation type (price / qty / unmatched)
- Expected vs actual values
- Deviation % and absolute
- Acknowledged (boolean)
- Acknowledged by (many2one)

MODIFIED MODELS

`account.move`

- Approval status field
- Approvals tab in form view
- Deviation summary & detail
- Follow-up widget
- Confirm button override
- Approve / Reject buttons

`purchase.order`

- Approval status field
- Approvals tab in form view
- Follow-up widget
- Confirm button override
- Approve / Reject buttons

`res.partner`

- Business Unit field (selection)

DELIVERY & RESPONSIBILITY

Odoo Custom Dev

- Approval workflow engine
- PO deviation checks
- Approval configuration views
- Centralized approval list
- Email templates (4)
- Security groups (3)

VVF Consulting

- Budget control module
- Manufacturing indirect cost allocation
- Budget ↔ approval integration

Sign-off sequence: Customer validates this spec → Odoo confirms feasibility → Odoo develops workflow → VVF delivers budget module in parallel.

Version Log

Document revision history

VERSION	DATE	AUTHOR	CHANGES
v1.0	March 2026	VVF Consulting	Initial spec — approval workflow for Vendor Bills & Purchase Orders (BU-based routing, sequential approval, notifications, follow-up UI)

v2.0	27 March 2026	VVF Consulting	• Added PO deviation control (threshold-based checks on price, qty, line total) • Added Budget Control Module scope (VVF-developed, linked to manufacturing indirect costs / IAS 2) • Updated Implementation Summary with responsibility split (Odoo vs VVF)